

What We Did, Simply

Uncertainty-aware visible–infrared ship detection: what we tested, what held up, and what did not
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Summary

- We built a ship and buoy detector that combines a visible camera with a thermal infrared camera.
- The central question was whether each detector’s predicted uncertainty could decide, frame by frame, which camera to trust.
- The uncertainty proved informative, but no way of using it made the combined system more accurate.
- A simple image-brightness rule worked better in daylight. At night it failed once the detector was retrained on corrected labels.

The problem

Maritime systems often pair a visible camera with a thermal infrared camera, because the two fail under different conditions. Visible imagery degrades in fog, glare and darkness. Thermal imagery loses vessels when they reach the same temperature as the surrounding water.

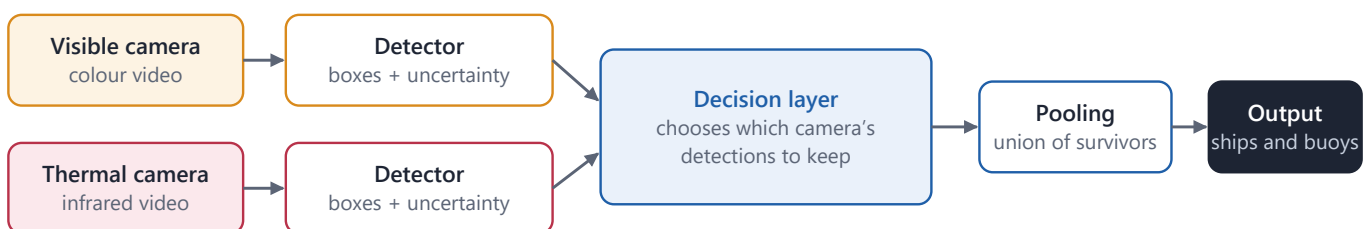
Neither failure is signalled. A degraded detector simply returns fewer detections, and a combining rule cannot tell that apart from an empty scene.

The question

Modern detectors can report, alongside each bounding box, an estimate of how uncertain that box is. We asked whether this predicted uncertainty is reliable enough to decide, frame by frame, how much weight each camera should receive.

What we built

- **Data:** the Pohang Canal dataset, real video recorded from a vessel in South Korea. It holds 158,319 images and 1.18 million annotated ships and buoys across five recordings.
- **Detectors:** one YOLO detector per camera, each extended with an output that estimates the uncertainty of every box.
- **Fusion:** a decision layer that selects which camera’s detections to keep, then pools the survivors.



How we kept the results honest

- **Pre-registration:** the success criteria for each major test were written down before the test was run.
- **A held-out recording:** one of the five recordings was kept untouched and evaluated exactly once, at the end.
- **Measured noise:** we measured how much scores vary by chance, so small differences were not mistaken for improvements.

Limitations

Fog, glare and low light were simulated on clean images rather than recorded. All data comes from a single location, and only one recording was made at night.

What we found

Accuracy is reported as average precision (**AP**), the standard detection score, from 0 (nothing found) to 1 (perfect).

1. Uncertainty is informative, but it does not improve the system

The test compared the real uncertainty values with the same values shuffled randomly across boxes. If the real values carry information, they should beat the shuffled ones.

- As a weight on box positions, real uncertainty beat the shuffled control in 0 of 4 conditions.
- As a way to re-rank which boxes to keep, it beat the control in 3 of 4, meeting the pre-registered threshold.
- However, a system that ignored uncertainty entirely scored higher in 6 of 8 conditions.

A further variant learned to combine uncertainty with the detector's confidence. It gave a small gain on all five retrained detectors, but only two cleared the margin fixed in advance, so we do not claim it.

2. A simple rule worked, until the detector changed

The rule: *at night, if the image is dark or foggy, discard the visible camera.*

- In daylight it outperformed the visible camera alone in all 4 conditions (+0.006 to +0.011 AP).
- At night it appeared justified, because the visible detector scored 0 at night.

That zero turned out to be an artefact. A label-filtering step had a bug and removed many ship annotations from the night training images, so the visible detector never learned to detect at night. After the labels were restored and the detector retrained, it scores 0.25 AP at night. The unchanged rule still discards it, and night accuracy falls to the thermal camera's 0.07.

Lesson: a rule for choosing between cameras is really a statement about the detectors behind them, and it must be re-checked whenever a detector is retrained.

3. Many small improvements were within the noise

Differences below roughly 0.003 AP cannot be distinguished from chance. Consecutive video frames are nearly identical, so standard error bars were at least 1.9 times too narrow. Re-examining 74 earlier findings with these corrections reclassified 20 as inconclusive.

4. The held-out test

On the untouched recording, the system scored 0.27 AP, which is 0.02 below its score on the development data. The uncertainty around that gap is too wide to say whether it is real.

Other approaches we tested

Approach	Outcome	Detail
Matching the same object across both cameras	Not useful	The cameras are not precisely aligned, so only 0.05% of visible-camera boxes found a thermal partner. In practice, fusion amounts to pooling both cameras' detections.
Selecting the best detector architecture (31 designs, 93 training runs)	No clear winner	The top three were statistically indistinguishable.
Refinements: uncertainty-weighted box merging, score recalibration, softer duplicate removal	Not adopted	None produced a clear improvement.

Where it stands

- The work is written up as a research paper that reports the negative results alongside the positive ones.
- Title: *Predicted Uncertainty Is Informative but Does Not Improve Visible-Infrared Maritime Detection.*
- Target venue: Transactions on Machine Learning Research (TMLR). The formatted LaTeX version compiles.
- Remaining: decide whether to shorten the main text, add acknowledgements, and submit.